lost altogether, can an ecosystem remain intact—or does this damage ecosystem function?

Such questions were the focus of the Biodiversity and Ecosystem Function (BEF) conference held in Paris in 2000. More than 60 leading international ecologists, including Michel Loreau, director of the Centre for Biodiversity Theory and Modeling in Moulis, France, outlined diverse research; some looked more closely at species, others at what makes an ecosystem work. Loreau maintains that a new unified ecological theory is necessary to combat extreme environmental challenges. That, he says, requires the integration of community ecology (the study of how species interact in ecosystems) with ecosystem ecology (research into the physical, chemical, and biological processes that connect organisms and their environment).

“Biodiversity loss… is likely to decrease the ability of ecosystems to resist the effects of climate change.”

Michel Loreau

Complex cycles

Scientists of both disciplines firmly believe that biodiversity, especially species and genetic diversity, is an important driver of ecosystem functioning. Ecosystems are powered by an input of energy and recycling of nutrients: plants and animals grow, die, and decompose, returning nutrients to the soil and restarting the cycle. These processes depend on the species within the ecosystems, which in turn depend upon one another as they interact—as predators and prey, for example. Many ecologists argue that a large variety of complementary species are needed to keep an ecosystem working and make it resilient to change. Others say that a few key species may be more important to stop ecosystems from collapsing.